

An Introduction to Faber Polynomials and Applications

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- From Chebyshev to Faber polynomials
- Definition of Faber polynomials: generating function and recurrence
- Example: the deltoid region ($m = 3$)
- Key properties
- Application: accelerating the power iteration
- Further applications and conclusion

Introduction

- **Chebyshev polynomials (of the first kind):**

$$T_n(x) = \cos(n \arccos(x))$$

They satisfy the following recurrence relation:

$$T_0(t) = 1, \quad T_1(t) = t, \quad T_{N+1}(t) = 2t T_N(t) - T_{N-1}(t), \quad N \geq 1.$$

- **Lemma.** For all $n \geq 0$, the Chebyshev polynomials satisfy

- ▶ $|T_n(x)| \leq 1$ for $x \in [-1, 1]$

- ▶ For all $\varepsilon > 0$,

$$|T_n(x)| \geq \frac{1}{2} \left(1 + \sqrt{2\varepsilon}\right)^n \quad \text{for } x \in \mathbb{R} : |x| = 1 + \varepsilon$$

- **Optimality:** A monic polynomial of degree n has the form $p_n(x) = x^n + \sum_{j=0}^{n-1} b_j x^j$
For any such p_n (see MAD6407),

$$\max_{x \in [-1, 1]} |p_n(x)| \geq 2^{1-n}$$

The scaled Chebyshev polynomial $\tilde{T}_n = 2^{1-n} T_n$ achieves this bound $\rightarrow$ useful for best uniform approximation (BUA) of functions

Note: 2^{n-1} is the leading coefficient of T_n

From Chebyshev to Faber Polynomials

- **Extension to $\mathbb{C}$:** The Chebyshev polynomials extend to the complex plane via

$$T_n(z) = \cosh(n \operatorname{arccosh}(z)), \quad z \in \mathbb{C}$$

They are bounded inside the **Bernstein ellipse**

$E_\rho = \left\{ \frac{1}{2}(\rho e^{i\theta} + \rho^{-1} e^{-i\theta}) : \theta \in [0, 2\pi] \right\}$ with foci ± 1 , and grow exponentially outside it.

- **Limitation:** The Bernstein ellipse is tied to the interval $[-1, 1]$

For a general compact set $K \subset \mathbb{C}$, there is **no reason** for its boundary to resemble a Bernstein ellipse.

- $\Rightarrow$ **Natural question:** given an **arbitrary** compact set $K \subset \mathbb{C}$, can we construct monic polynomials that are **optimally small** on K , just as $\tilde{T}_n$ is optimally small on $[-1, 1]$?

Yes: these are the Faber polynomials $\rightarrow$ defined via the conformal map of the exterior of K onto the exterior of the unit disk.

Definition of Faber Polynomials

- **Setup:** Let $\Omega \subset \mathbb{C}$ be a compact set whose complement $\mathbb{C} \setminus \Omega$ is simply connected. By the **Riemann mapping theorem**, there exists a conformal map

$$\psi : \{w \in \mathbb{C} : |w| > 1\} \rightarrow \mathbb{C} \setminus \Omega$$

with $\psi(\infty) = \infty$ and $\psi'(\infty) > 0$.

- **Definition:** The Faber polynomials $P_n(z)$, $n \geq 0$, for Ω are defined by the generating function

$$\frac{\psi'(w)}{\psi(w) - z} = \sum_{k=0}^{\infty} F_k(z) w^{-k-1}, \quad |w| > 1, z \in \Omega$$

- **Key property:** When ψ is rational, the Faber polynomials satisfy a **recurrence relation** $\rightarrow$ efficient to compute.

Faber Polynomials: From Generating Function to Recurrence

- **Hypocycloid domains:** For an m -cusped hypocycloid with conformal map

$$\psi(w) = \frac{m-1}{m}w + \frac{1}{m}w^{1-m}$$

the Faber polynomials P_n satisfy

$$P_{n+1}^{(m)}(z) = \frac{m}{m-1}zP_n^{(m)}(z) - \frac{1}{m-1}P_{n-m+1}^{(m)}(z), \quad n \geq m-1$$

with enclosing curve

$$\gamma^{(m)}(t) = \frac{m-1}{m}e^{it} + \frac{1}{m}e^{(1-m)it}, \quad t \in [0, 2\pi]$$

- **Special cases:**

- ▶ $m=2$: $\psi(w) = \frac{1}{2}(w + w^{-1})$, enclosing curve is $[-1, 1] \rightarrow$ recovers the [Chebyshev polynomials](#) T_n
- ▶ $m=3$: gives a [deltoid recurrence](#) $\rightarrow$ next slide

Faber Polynomials on a Deltoid Region ($m = 3$)

- **Conformal map:**

$$\psi(w) = \frac{2}{3}w + \frac{1}{3}w^{-2}$$

maps the exterior of the unit disk onto $\mathbb{C} \setminus \Gamma$.

- **Boundary curve** Consider the deltoid curve $\gamma \subset \mathbb{C}$ parameterized by

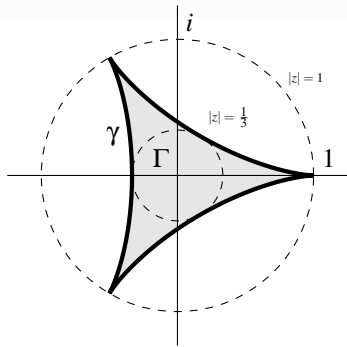
$$\gamma(t) = \frac{2}{3}e^{it} + \frac{1}{3}e^{-2it}, \quad t \in [0, 2\pi]$$

and let Γ be the closure of the region enclosed by γ .

- **Recurrence:** For $n \geq 2$,

$$P_{n+1}(z) = \frac{3}{2}zP_n(z) - \frac{1}{2}P_{n-2}(z)$$

with $P_0(z) = 1$, $P_1(z) = z$, $P_2(z) = z^2$.



The curve γ , called a deltoid, intersects the circle $|z| = 1$ at three points and $|z| = \frac{1}{3}$ at three points. The closure of the region enclosed by γ is Γ .

P. Cowal, N. F. Marshall, S. Pollock, Faber polynomials in a deltoid region and power iteration momentum methods

Faber Polynomials: Visual Representation

- Plots of $|P_n(z)|$ on $\{z \in \mathbb{C} : |\operatorname{Re}(z)| \leq 1, |\operatorname{Im}(z)| \leq 1\}$
- Points with $|P_n(z)| \leq 10^{-3}$ shown in black, points with $|P_n(z)| \geq 1$ shown in white
- As n grows, the region where $|P_n| \leq 1$ increasingly resembles the **deltoid region** Γ

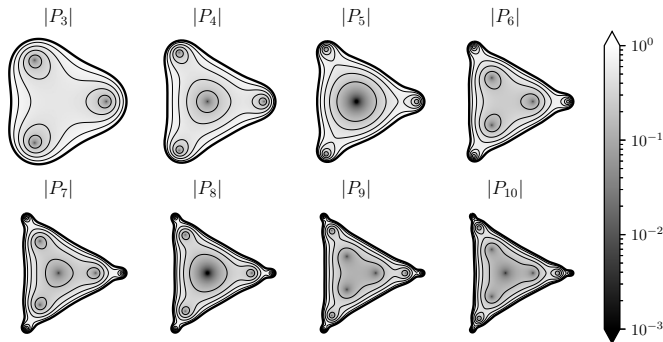


Figure 2, P. Cowal, N. F. Marshall, S. Pollock, *Faber polynomials in a deltoid region and power iteration momentum methods*,

2025

Key Properties of Faber Polynomials

- **Monic:** The Faber polynomial $P_n(z)$ is monic of degree n , i.e.,

$$P_n(z) = z^n + \text{lower order terms}$$

- **Optimality:** Among all monic polynomials of degree n , the Faber polynomial P_n has the smallest uniform norm on the boundary ∂K

$$\min_{\substack{p \text{ monic} \\ \deg p = n}} \max_{z \in \partial K} |p(z)| = \text{cap}(K)^n$$

where $\text{cap}(K)$ is the **logarithmic capacity*** of $K \rightarrow$ a measure used in **potential theory**

(analogous to the Chebyshev optimality mentioned before: $\tilde{T}_n = 2^{1-n}T_n$ has the smallest uniform norm on $[-1, 1]$ among all monic polynomials of degree n)

- **Zeros:** The zeros of P_n are all contained inside K , analogous to the zeros of T_n lying in $[-1, 1]$ M. X. He, E. B. Saff, *The zeros of Faber polynomials for an m -cusped hypocycloid*, Journal of Approximation Theory, 1994

* P. K. Suetin, *Series of Faber Polynomials*, 1998

Application: Accelerating the Power Iteration (I of III)

- **Power iteration:** Starting from x_0 , repeatedly apply

$$x_k = u_k / \|u_k\|, \quad u_{k+1} = Ax_k$$

converges to the dominant eigenpair (λ_1, ϕ_1) at rate $|\lambda_2/\lambda_1| \rightarrow$ **slow when**
 $|\lambda_2/\lambda_1| \approx 1$

- **Key idea:** instead of iterating with A^N , apply a **polynomial** $p_N(A)$ generated by a recursion:

$$u_{k+1} = Ax_k - \beta h_k^{-1} x_{k-1}, \quad h_k = \|u_k\|, \quad x_k = h_k^{-1} u_k$$

where x_k approximates the **dominant eigenvector** ϕ_1 and β is a parameter to be chosen.

- ▶ Requires only one matrix-vector product per step
 - ▶ **Converges faster** if β is chosen optimally
 - ▶ **Challenge:** the optimal β depends on λ_2 , which is unknown!
- **Limitation of Chebyshev:** For **complex eigenvalues**, $T_N(z)$ grows exponentially outside $[-1, 1]$ for $z \in \mathbb{C}$ off the real line, so Chebyshev is no longer the right filter.

Application: Accelerating the Power Iteration (II of III)

- **Faber polynomial iteration ($m = 3$):** For matrices whose non-dominant eigenvalues lie in $\lambda_2\Gamma$ (here $|\lambda_1| > |\lambda_2| \geq \dots \geq |\lambda_n|$), use

$$u_{k+1} = Ax_k - \frac{\beta}{h_k h_{k-1}} x_{k-2}, \quad h_k = \|u_k\|, \quad x_k = h_k^{-1} u_k$$

with optimal parameter $\beta = 4\lambda_2^3/27$

- **Why does it work?** The Faber polynomials P_N satisfy

$$\triangleright |P_N(z)| \leq 1 \quad \text{for } z \in \Gamma$$

$$\triangleright |P_N(z)| \geq \frac{1}{3}(1 + \sqrt{\varepsilon})^N \quad \text{for } z \in \mathbb{C} : |z| = 1 + \varepsilon$$

so the dominant eigenvalue (λ_1) grows while all others stay bounded.

- **Square root gain over the power method*:** This new accelerated iteration needs only $\sim \sqrt{N}$ **steps** to achieve the same accuracy

$$\underbrace{(1 + \sqrt{\varepsilon})^{-N}}_{\text{Faber}} \ll \underbrace{(1 + \varepsilon)^{-N}}_{\text{Power method}}, \quad \varepsilon = |\lambda_1/\lambda_2| - 1 \text{ small}^*$$

*Theorem 4, P. Cowal, N. F. Marshall, S. Pollock, Faber polynomials in a deltoid region and power iteration momentum methods

Application: Accelerating the Power Iteration (III of III)

- **Challenge:** the optimal parameter $\beta = 4\lambda_2^3/27$ requires knowledge of λ_2 , which is **unknown in practice**.
- **Dynamic implementation:** approximate $\beta_k = 4(v_k r_k)^3/27$ using *a posteriori* information:
 - ▶ **Rayleigh quotient:** $v_k = v_{k+1}^T x_k \approx \lambda_1$
 - ▶ If we had chosen β_k to attain the optimal convergence rate $\rho = e^{-\sqrt{1/r-1}}$, where $r = |\lambda_2/\lambda_1|$, then solving for r in terms of ρ we have

$$r = \frac{1}{(\log \rho)^2 + 1}$$

- ▶ The residual convergence rate ρ_k is **computable**, so we approximate

$$r_{k+1} = \frac{1}{(\log \rho_k)^2 + 1} \approx \left| \frac{\lambda_2}{\lambda_1} \right|$$

- **Result:** even without knowing λ_2 , the dynamic method achieves a similar convergence rate to the static optimal method*

* P. Cowal, N. F. Marshall, S. Pollock, Faber polynomials in a deltoid region and power iteration momentum methods

More Applications of Faber Polynomials

- **Solving linear systems:** Faber polynomials are used in hybrid iterative methods for **nonsymmetric** linear systems M. Eiermann, X. Li, R. S. Varga, *On hybrid semi-iterative methods*, 1989
- **Matrix functions:** approximating $f(A)$ (e.g., the matrix exponential e^A) via truncated Faber series I. Moret, P. Novati, *The computation of functions of matrices by truncated Faber series*, 2001
- **Eigenvalue problems:** Arnoldi-Faber methods for large non-Hermitian eigenvalue problems V. Heuveline, M. Sadkane, *Arnoldi-Faber method for large non Hermitian eigenvalue problems*, 1996
- **Higher order generalizations:** arbitrary m -cusped hypocycloid regions via higher order recurrences P. Cowal, N. F. Marshall, S. Pollock, *Random walks, Faber polynomials, and accelerated power methods*, 2025

- **Faber polynomials** generalize Chebyshev polynomials to general compact sets $K \subset \mathbb{C}$, naturally adapted to the geometry of the domain via a conformal map ψ
- When ψ is rational, Faber polynomials satisfy a **recurrence relation**, making them efficient to compute and naturally associated to **hypocycloid regions**
- **Application (of many more applications)**: for matrices with complex eigenvalues in a deltoid region, Faber polynomial filters achieve a **square root gain** over the power method!

Thanks for your attention!